Garrett J. Kepler

Github | Academic Website | Email

Education

Fall 2022 - Present	**Washington State University (WSU)** Ph.D. in Mathematics Research Topic: Spectral Graph Theory	(GPA: 3.55/4.0)
Fall 2023 - Spring 2026	**WSU** M.S. in Statistics Research Topic: Random Sampling & Clustered Networks	(GPA: 3.55/4.0)
Spring 2019 - Spring 2022	**Cal State University - East Bay** B.S. in Mathematics with Minor in Physics	(GPA: 3.412/4.0)

Teaching Experience

Graduate Teaching Assistant — August 2022 - present

In WSU's Ph.D. program, I have taught the following mathematics courses through the either as a Teaching Assistant (TA) or Instructor of Record (IOR):

- Math 300 (current): Mathematical Computing (IOR)
- Math 325: Combinatorics (IOR)
- Math 315: Differential Equations (IOR)
- Math 273: Calculus III (IOR)
- Math 220: Linear Algebra (IOR)
- Math 172: Calculus II (TA)
- Math 171: Calculus I (IOR)
- Math 108: Trigonometry (IOR)
- Math 103 (Accelerated & Regular): Algebra Methods and Introduction to Functions (TA)
- Math 100 (Accelerated): Basic Mathematics (IOR/TA)

Graduate Tutor — August 2022 - present

WSU's Math Learning Center offers free tutoring to all currently enrolled students. As a graduate tutor, I am expected to be proficient in every course offered and be able to assist in every topic. Semester weekly assignments have ranged from 4-8 hours per week.

Private Tutor — Current

I tutor privately to both high school and undergraduate students in the Pullman area and beyond. I have mostly focused on Calculus, Data Science, Algebra, and Spanish when tutoring. About 4 hours per week.

Athletics Tutor — August 2023-January 2024

WSU's Athletics Department hosts their own tutoring center specifically for student athletes. I was expected to assist 2-4 students every week with hours ranging from 6-12 hours per week. Topics included both mathematics and statistics.

Research Experience

Local Limits of Random Graphs (Summer School) Link to Program Website

I attended a summer school in 2025 funded by the NSF through Simon Laufer Mathematical Sciences Institute and the Fondation Mathématique Jacques Hadamard. Defining a specific topology over graphs, we can use probability theoretic techniques to investigate convergence of graph sequences. Most importantly, we can take into account local geometry under this topology. This way we can infer information about the limiting geometry as well as the sequence of graphs that approach the limiting object.

Research on the Mathematics of the Bay Area (RUMBA) Link to Program Website

Founded by Dr. Andrea Arauza Rivera, RUMBA serves as an opportunity for undergraduates to get hands-on experience in data science and mathematics. The projects I was involved in focus on inequality in the Bay Area. Firstly, an assessment of correlation between community factors and standardized test scores. Secondly, an analysis of food access within Bay Area counties.

Graduate Collaborations

In my current Ph.D. program, we are encouraged to collaborate within our department and with other departments as well. I currently have ongoing projects with two fellow mathematics graduate students as well as a chemical engineering student. The projects surround Ramsey Theory, eigenvectors of graphs, spectral-based redistricting methods, and applications of machine learning in catalysis respectively. Past project collaborations have been in inverse eigenvalue problems, data science, statistics, and computer science (see below).

Research Publications

Kepler, Garrett et al. (Mar. 2023). "Food Deserts and k-Means Clustering". In: *SIURO* 16. URL: https://www.siam.org/publications/siam-journals/siam-undergraduate-research-online-siuro/issues/siuro-volume-16/.

Brannan, Jared J. L. et al. (Aug. 2024). "Properties of the cone of polynomials of fixed degree that preserve nonnegative matrices". In: URL: https://arxiv.org/abs/2402.04508.

Briscoe, Jarren et al. (May 2025). "Algorithmic Accountability in Small Data: Sample-Size-Induced Bias Within Classification Metrics". In: *AISTATS*. DOI: 10.48550/arXiv.2505.03992.

Academic Community Involvement

- Algebra and Discrete Math Seminar Organizer (2025-2026 WSU)
- WNAR of IBS session chair (2026) "Clinical Trials & Drug Development"
- WNAR of IBS conference talk (2026) "A Machine Learning Approach to Catalysis Modeling"
- Guest Lectures:
  - Graph Theory (2026 WSU) "At the Intersection of Graph Theory & Probability Theory"
  - Mathematical Snapshots (2026 WSU) "A Random Talk on Random Walks"
  - Graph Theory (2025 WSU) "Applications in and Methods of Spectral Graph Theory"
  - Advanced Linear Algebra (2025 WSU) "MCMC: An Introduction"
  - Mathematical Snapshots (2025 WSU) "Graph Theory Applications: Gerrymandering"
- Graduate Outreach Organization in Physics Colloquium (2026) "A Science Chit Chat by a Mathematician"

- California State East Bay Mathematics Colloquium Speaker (2025) "Graph Theory and Gerrymandering: Computationally Assessing Fairness"
- Joint Mathematics Meeting conference talk (2025) "Eigenspaces of Graphs and their Utility"
- WSU Seminar on Theory and Applications of Discrete Math, Linear Algebra, and Number Theory:
  - "Graphgebrability & Mathematical Exploration" (2026)
  - "A Look into Limits of Random Graphs" (2025)
  - "A Probabilistic Excursion in Spectral Graph Theory" (2025)
  - "Graphical Inginuety: Spectral Solutions to Combinatorial Conondrums" (2024)
  - "Laplace, Fiedler, & Markov: A Graph Reconstruction Problem" (2024)
- Western Canadian Linear Algebra Meeting conference talk (2024) "Independent Sets and Graph Energy"

Mentoring

Undergraduate Research Project Mentoring

American Women in Mathematics (AWM) Mentoring Program

WSU's AWM chapter started a mentoring program in August 2025. Pairing graduate students with undergraduate students, we aim to demystify collegiate, research, and industry experiences as well as expose undergraduate students to opportunities they might otherwise miss. My mentee Steven Brand and I meet regularly to discuss these topics over coffee from August 2025 on.

High School Project Mentoring

From August 2024 to August 2025, a fellow graduate student and I mentored a proactive high school mathematician in research. Kabir Shah was interested mostly in the applications of graph thoery. We guided him into proving fun theorems and creating a write-up of interesting application areas.

Awards

- "Nancy J Robertson Endowed Graduate Fellowship in Mathematics" Washington State University 2026
- "High Acheivement in Mathematics Research & Collaboration" California State East Bay 2022